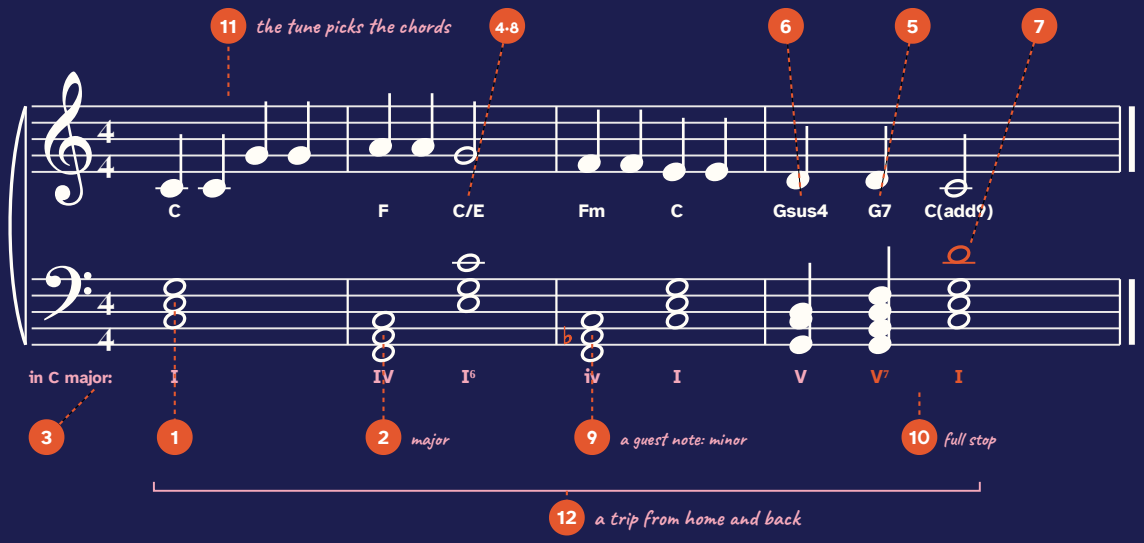
PART IV · WHICH NOTES SOUND TOGETHER

Harmony

A melody is the singer; harmony is the band behind it: the chords underneath. Every chapter of this part is in these four bars of Twinkle.



1 **Chords and triads** p. 67
three notes stacked like a snowman

2 **Four kinds of triad** p. 70
major 3 + 2, minor 2 + 3

3 **Chords in a key** p. 72
Roman numerals: I, IV and V do most of the work

4 **Inversions** p. 74
the same notes, a different one at the bottom

5 **Seventh chords** p. 76
one more floor: G7 pulls home

6 **Colour chords** p. 78
sus and add: a note moved or added

7 **The tower: 9, 11, 13** p. 80
keep skipping a letter

8 **Slash chords** p. 82
C/E: the chord, then the bass note

9 **Chords from outside the key** p. 84
a guest note: A♭ in C major

10 **Cadences** p. 87
V to I ends a phrase like a full stop

11 **Chords under a melody** p. 89
the tune's main notes pick the chord

12 **Progressions everywhere** p. 91
the same few trips, in every song

★ **Tips: when a chord sounds wrong** p. 93

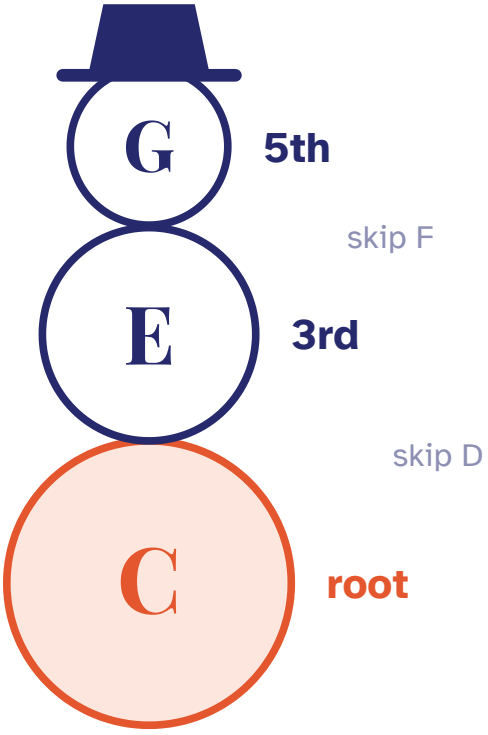
PART IV · HARMONY · 1

Chords and triads

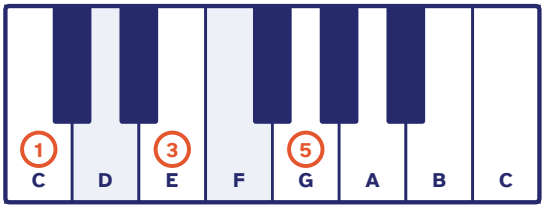
Press two or more notes at once and you have a **chord**. The favourite of all is the **triad**: three notes stacked like a snowman.

THE SECRET

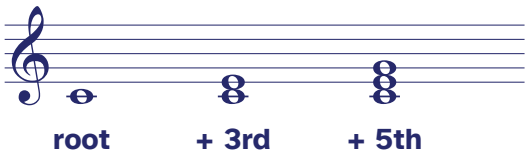
Play one, skip one, play one: the snowman rule.



1. Pick a note: the **root**, the big ball at the bottom. It holds the chord up and gives it its name.
2. Skip one letter, take the next: **E**, the **3rd**.
3. Skip again: **G**, the **5th**. C E G is the triad of C.



On the keys: play one, skip one (shaded), play one, skip one, play one.



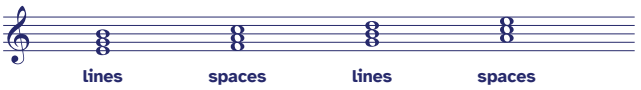
The snowman on the staff: three lines in a row (or three spaces).

On the staff, each ball sits on the **next line up** (or the next space up): the root, a **3rd** above it, a **5th** above the root.

First met: Level ½ (the tonic triad of C, G and F major).

All lines or all spaces

A stacked triad sits on **three lines in a row** or in **three spaces in a row**, in a neat stack. If one note breaks the pattern, it is not a triad.



Lines, spaces, lines, spaces: every one a triad.

PART IV · HARMONY · 1 · CONTINUED

Solid and broken

Played together, a chord is **solid** (or blocked). Played one note at a time, it is **broken**. Spread up and down the keyboard, a broken chord becomes an **arpeggio**. Left-hand accompaniments, the arpeggios in technical work and the chord names over a song are all triads.

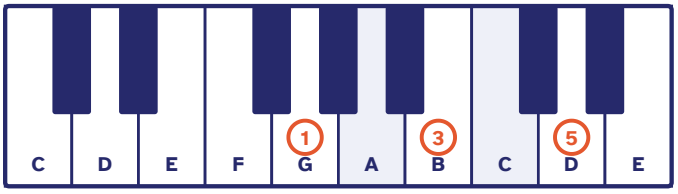
First met: Level 0; arpeggios from Level 1.



The same three notes, solid then broken.

EXAMPLE 1

Build a triad on G



1. **Start on G:** the root.
2. **Skip A, play B:** the 3rd.
3. **Skip C, play D:** the 5th.

G B D: the triad of **G**. Same shape as C E G, just moved up.

EXAMPLE 2

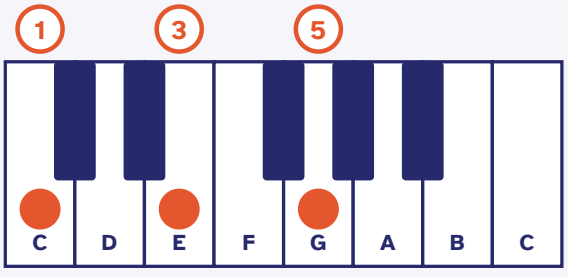
Is it a triad?



1. **Chord 1:** D F A, all in spaces, each a 3rd apart: **a triad**.
2. **Chord 2:** D E A: D to E is a 2nd. **Not a triad.**
3. **Chord 3:** D G C, stacked in 4ths. **Not a triad.**

Only chord 1 is a triad. Check that each note is a **3rd** above the one below.

TRY IT · 10 SECONDS



Thumb on C. Fingers 1, 3, 5 down together.

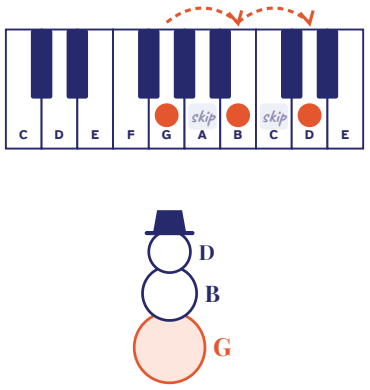
That is the C chord. Slide your hand to any white key and do it again: the snowman shape always works.

PART IV · HARMONY · 1 · CONTINUED

GAME

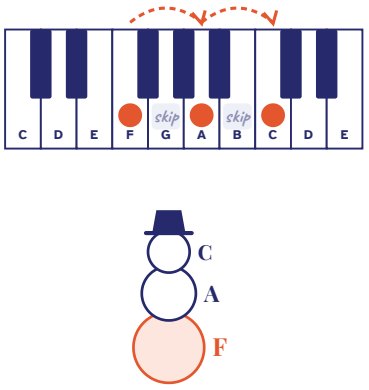
Build one yourself

G



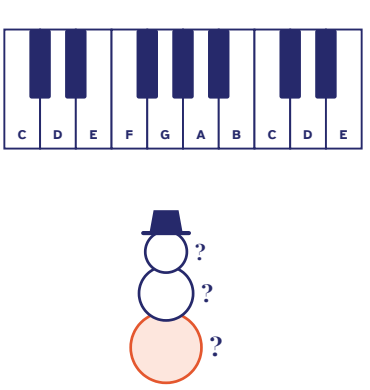
Start on G: **G B D**.

F



Start on F: **F A C**.
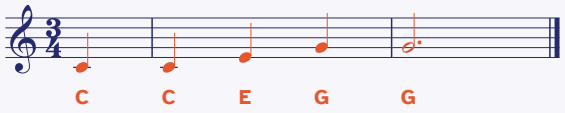
?



Your turn: start on **A**. Which three keys?

Build the snowman on A: play one, skip one, play one, skip one, play one.

Answer: A C E, the A minor chord.

YOU KNOW THIS

The Blue Danube (Johann Strauss II), here in C



The most famous waltz tune simply climbs the **C triad**: **C, C E G, G**. Root, 3rd, 5th, then it holds the top: that is why it sounds so open.

Try it. Play C E G one at a time, then hum the waltz over it.

THINK OF IT Solid: three friends shout “Hello!” together. Broken: each says it in turn.

TRAP

- × Stacking 2nds or 4ths: a triad skips exactly one letter each time.
- × Naming a chord from its top note: the root names it (at the bottom when the notes are stacked in 3rds).

SEE ALSO
Counting intervals p. 51 · Major scales p. 55

PRACTISE IT
Workbook Harmony 1 · Online: Library › Chords

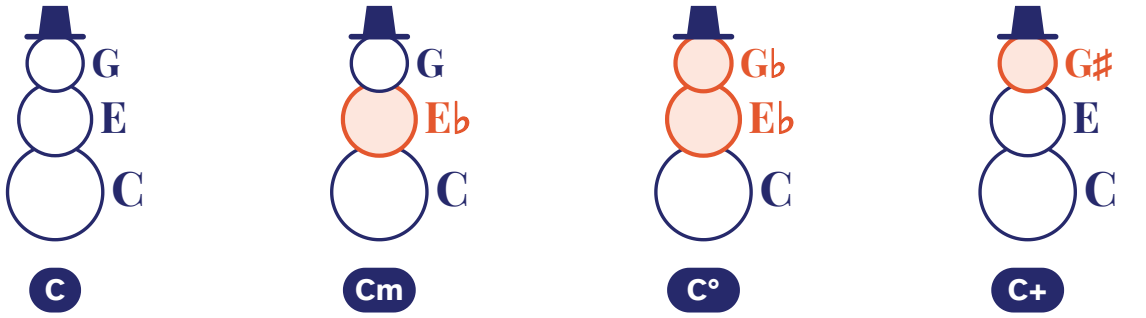
PART IV · HARMONY · 2

Four kinds of triad

Every triad is two 3rds stacked. To tell the kinds apart, count the keys **in between**, black and white, that you do not play: root to 3rd, then 3rd to 5th. Those two numbers name the kind, and its sound.

THE SECRET

Count the keys between: 3 + 2 is major, 2 + 3 is minor.

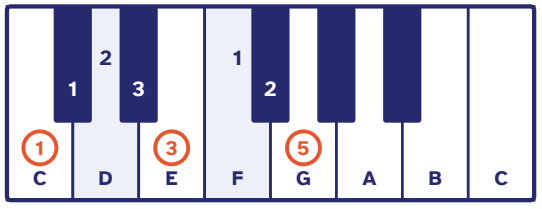


Count the keys between: 3 + 2 is major, 2 + 3 is minor

KIND	KEYS BETWEEN	ON C	CHORD NAME	FIRST MET
major	**3 + 2**	C E G	C	Level ½
minor	**2 + 3**	C E♭ G	Cm	Level 2
diminished	**2 + 2**	C E♭ G♭	C° or Cdim	Level 5
augmented	**3 + 3**	C E G♯	C+ or Caug	Level 6

Chord names: a letter alone means major (C); m means minor (Cm). You see them above the melody in songbooks and lead sheets: C | Am | F | G. In interval names, 3 keys between is a major 3rd and 2 is a minor 3rd (p. {S2}).

Count the keys in between



Play C and E and count the keys you skip, black and white: C♯, D, D♯ is **3**. Then E to G: F, F♯ is **2**. **3 + 2 is major.** Any root works the same way: from D, skip 3 to F♯, then 2 to A: D F♯ A, D major.

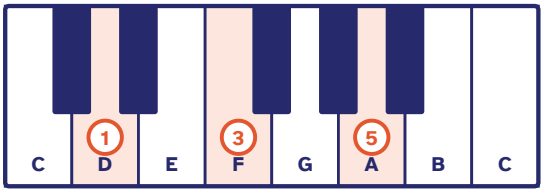
PART IV · HARMONY · 2 · CONTINUED

How they sound

KIND	SOUNDS	HEARD IN
major	bright, settled, open	most cheerful tunes, fanfares
minor	darker, softer, sadder	lullabies, laments, film tension
diminished	tense, unstable, wants to move	suspense, the chord before home
augmented	floating, dreamy, unsettled	dream scenes, a passing colour

EXAMPLE

Major or minor? D F A



1 **Root to 3rd:** D to F skips D♯ and E: **2**.
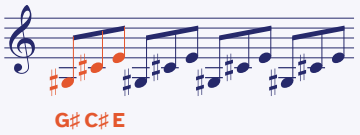
2 **3rd to 5th:** F to A skips F♯, G and G♯: **3**.

3 **2 + 3** is minor.

D F A is **D minor** (Dm). Three of the white-key triads are minor: D, E and A.

YOU KNOW THIS

Beethoven, Moonlight Sonata (first movement)



The famous slow triplets are the notes of a **C♯ minor triad**: C♯, E and G♯, rocking gently. Minor is why it sounds so dark and sad.

THINK OF IT Two floors of a building: a tall floor (3 keys between) and a short one (2). Major: tall, then short, 3 + 2. Minor: short, then tall, 2 + 3. Diminished: two short, 2 + 2. Augmented: two tall, 3 + 3.

TRAP

- × Counting the keys you play: count only the keys in between, black and white.
- × Forgetting that in diminished and augmented triads the 5th changes too.
- × Spelling with the wrong letter: a triad skips one letter, so C minor is C E♭ G, never C D♯ G.

SEE ALSO
Chords and triads p. 67 · Major, minor and perfect p. 53

PRACTISE IT
Workbook Harmony 2 · Online: Library › Chords

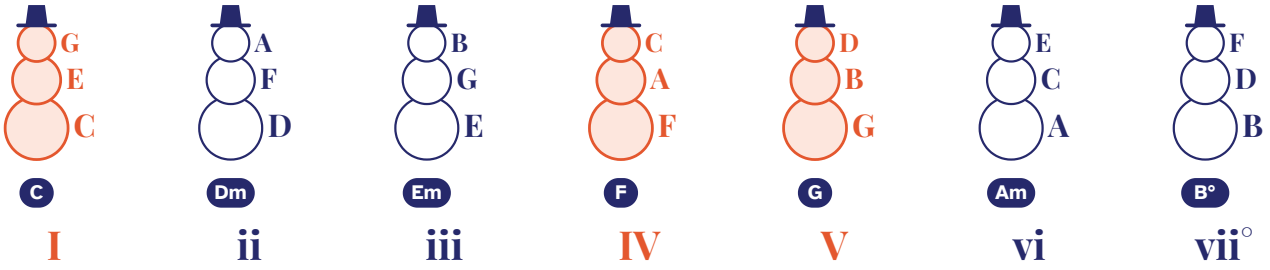
PART IV · HARMONY · 3

Chords in a key
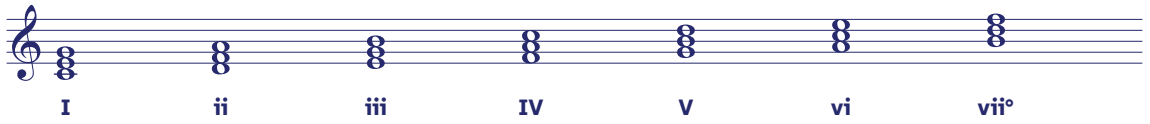
Build a triad on every note of a scale and you have the chords of that key. **Roman numerals** name each one by its place in the scale: I on the 1st note, IV on the 4th, V on the 5th. Capitals are major, small numerals are minor, and small with ° is diminished.

THE SECRET

Build a snowman on every note of the scale: I, IV and V are the strongest.



One snowman on every note: I, IV and V do most of the work



The seven chords of C major. Only white keys, so the kinds come out as on p. 71.

NUMERAL	NAME	KIND	KEYS BETWEEN	IN C	IN G
I	tonic	major	3 + 2	C	G
ii	supertonic	minor	2 + 3	Dm	Am
iii	mediant	minor	2 + 3	Em	Bm
IV	subdominant	major	3 + 2	F	C
V	dominant	major	3 + 2	G	D
vi	submediant	minor	2 + 3	Am	Em
vii°	leading note	diminished	2 + 2	B°	F♯°

The same in every major key: I, IV and V are major; ii, iii and vi are minor; vii° is diminished. First met: I, IV and V at Levels 1–2; the other chords from about Level 4.

The three primary chords
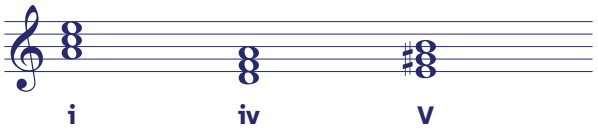
I, IV and V are the strongest chords of a key. Between them they hold **every note of the scale**: C E G, F A C, G B D. That is why so many tunes need only these three.

THINK OF IT I is home. IV is a visit next door. V is the walk back, already looking at the front door: it wants to go home to I. Play C, F, G, C and hear it.

PART IV · HARMONY · 3 · CONTINUED

In a minor key

In A minor the three main chords are **i** (A minor), **iv** (D minor) and **V**: E G♯ B, made major by the **raised 7th**, G♯, which pulls up to the home note A. Check: E G♯ B is 3 + 2, major.



A minor: i, iv and V. The G♯ makes V major.

Which chord holds the note?

NOTE IN C	C	D	E	F	G	A	B
Chord	I or IV	V	I	IV	I or V	IV	V

The same idea works in every key: find which of I, IV and V holds the note.

EXAMPLE

Find I, IV and V in G major



1. **Count up the scale:** G A B C D. The 1st note is G, the 4th C, the 5th D.
2. **Build a triad on each,** using the key signature: the F is F♯.
3. **Check with the keys between:** all three are 3 + 2, major.

In G major: **I = G**, **IV = C**, **V = D** (D F♯ A).

YOU KNOW THIS

The twelve-bar blues

I C	I C	I C	I C
IV F	IV F	I C	I C
V G	IV F	I C	V G

Almost every blues, and a lot of rock 'n' roll, is just **I, IV and V** in twelve bars. Learn this chart and you can play along with hundreds of songs.

Try it. Play the chart in C with your left hand, one chord per bar, four beats each.

TRAP

- × Writing every numeral big: ii, iii and vi are minor, so they are written small.
- × Forgetting the raised 7th in a minor key: V in A minor is E G♯ B, not E G B.
- × Forgetting the raised 7th in a minor key: in A minor, V is E major, with G♯.

SEE ALSO

Four kinds of triad p. 70 · Major scales p. 55 · Cadences p. 87

PRACTISE IT

Workbook Harmony 3 · Online: Library › Chords

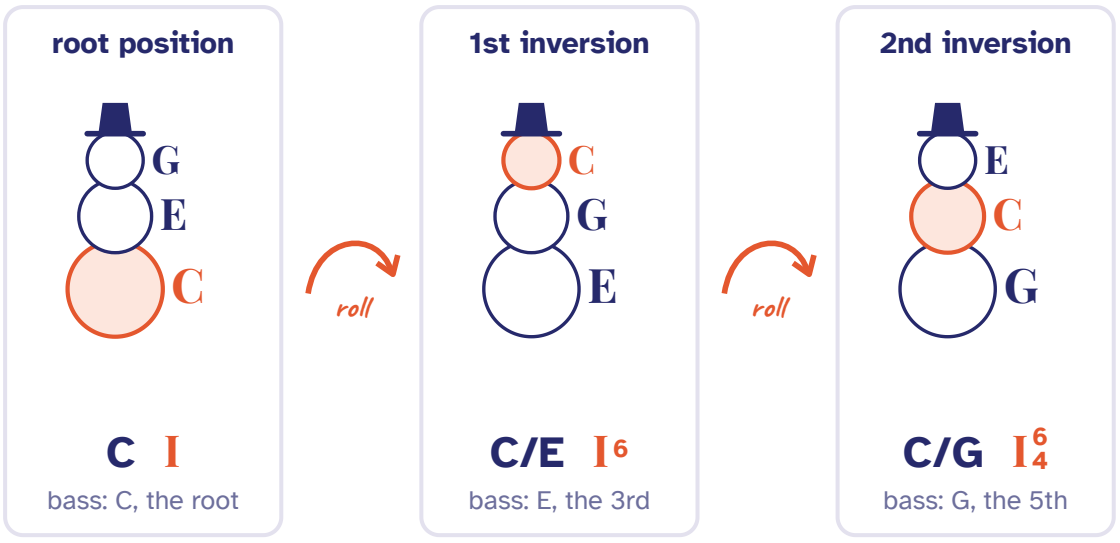
PART IV · HARMONY · 4

Inversions

Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red ball is the root, C: watch it travel.

THE SECRET

Move the bottom note to the top: same chord, new bass.
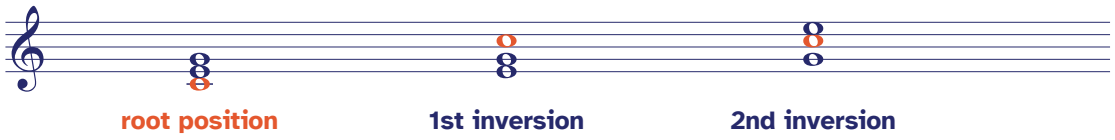


1. Start with the snowman **C E G**: root at the bottom.
2. Roll the bottom ball to the top: **E G C**. That is the 1st inversion.
3. Roll again: **G C E**, the 2nd inversion. Still **C major**.

Two ways to write it

	ROOT POSITION	1ST INVERSION	2ND INVERSION
Chord symbol pop, jazz	C	C/E	C/G
Harmony class numerals	I	I^6	I^6_4
Also written	Ia, 5/3	Ib, 6/3	Ic, 6/4
First met	Level ½	Level 3	Level 5

The little numbers count up from the bass. E G C: C is a **6th** above E, so I^6. G C E: E is a 6th and C a **4th** above G, so I^6_4. A slash chord just names the bass: C/E (more on p. 82).



The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom ball is the root.

PART IV · HARMONY · 4 · CONTINUED

Why inversions?

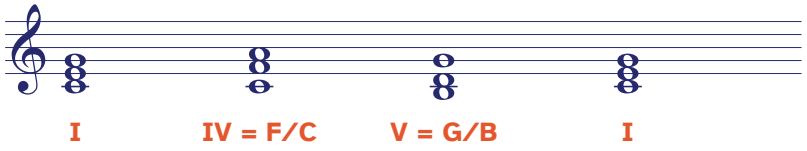
They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

Chords of C, every position

CHORD	ROOT POSITION	1ST INVERSION	2ND INVERSION
I (C)	C E G	E G C	G C E
IV (F)	F A C	A C F	C F A
V (G)	G B D	B D G	D G B
vi (Am)	A C E	C E A	E A C

EXAMPLE

Play I, IV, V, I smoothly
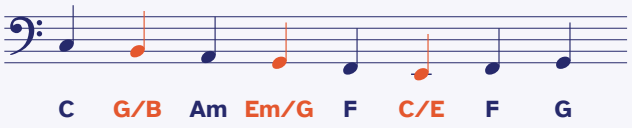


1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

IV is F/C and V is G/B: **inversions** keep the hand in one place.

YOU KNOW THIS

Pachelbel's Canon, the bass (here in C)



The famous bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G/B, Em/G and C/E put a different note in the bass.

Try it. Play the bass slowly and hum the famous violin tune over it.

THINK OF IT The same three friends in a photo, swapping places: still the same group (the same chord), but whoever stands at the front (the bass) changes the picture.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 67 · Chords in a key p. 72 · Cadences p. 87

PRACTISE IT

Workbook Harmony 4 · Online: Library › Chords

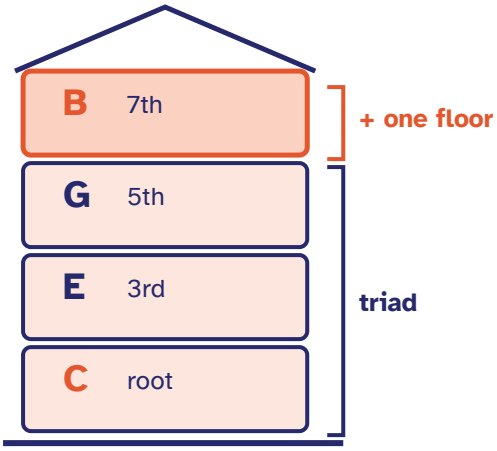
PART IV · HARMONY · 5

Seventh chords

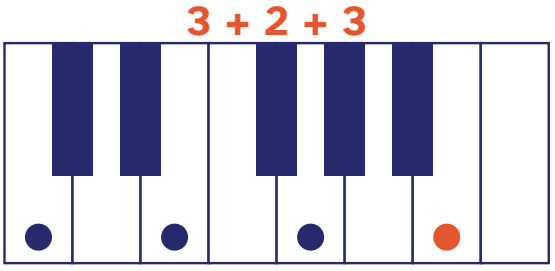
Add one more floor to a triad and the chord changes colour: dreamy, bluesy, spooky. Same rule as before: skip one note, stack the next. The new note is the **7th**.

THE SECRET

One more floor on the snowman and the chord changes colour.

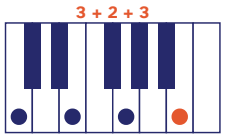


1. Build the triad: **C E G**.
2. Skip one, add the next: **B**.
3. Count the keys between: **3 + 2 + 3**. That is **Cmaj7**.



Five flavours, all on C

Cmaj7

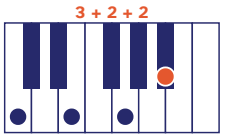


C E G B

dreamy

bossa nova, film

C7

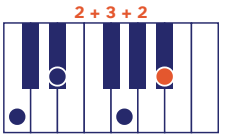


C E G B♭

bluesy

the blues

Cm7

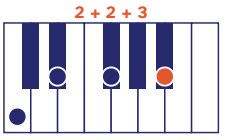


C E♭ G B♭

mellow

soul, jazz ballads

Cø7

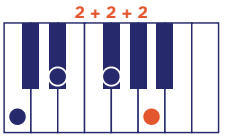


C E♭ G♭ B♭

searching

Bø7 is ii in A minor

C°7
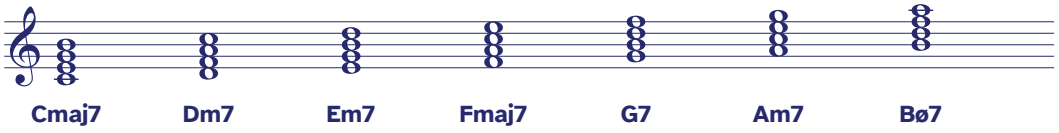


C E♭ G♭ B𝄫 (= A)

spooky

old film villains

How the names work: major triad + major 7th = **maj7** · major triad + minor 7th = **7** · minor triad + minor 7th = **m7** · diminished triad + minor 7th = **ø7** · diminished triad + diminished 7th = **°7**. Also written: CM7 or CΔ = Cmaj7, C–7 = Cm7, Cm7♭5 = Cø7, Cdim7 = C°7.

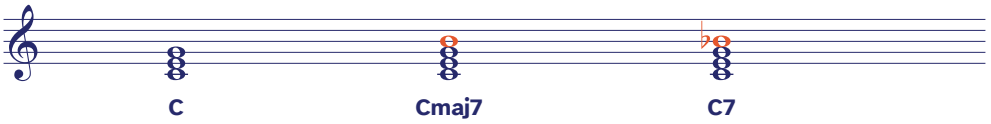
A seventh on every floor of C major



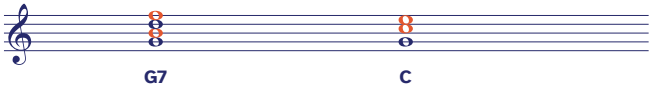
Only white keys. G7 is the one with the strongest pull home.

PART IV · HARMONY · 5 · CONTINUED

Spot the difference



One note decides: B makes Cmaj7, B♭ makes C7. Everything below stays the same.



Why G7 pulls home: its B wants to rise to **C** and its F wants to fall to **E**. Two small steps, and you are home in C.

EXAMPLE

Which seventh is it? G B D F



1. **Count the keys between:** G to B is 3, B to D is 2, D to F is 2.
2. **3 + 2 + 2** is the bluesy one.
3. **The root is G:** G7.

G B D F is **G7**, the chord that pulls hardest back to C.

YOU KNOW THIS

The blues, with sevenths

I C7	I C7	I C7	I C7
IV F7	IV F7	I C7	I C7
V G7	IV F7	I C7	V G7

Swap the triads for **7th chords** and the twelve-bar chart (p. 73) turns into the real blues: the 7th is the blue note.

Try it. Play the chart with plain C, F and G first (p. 73), then add the red 7ths: B♭, E♭ and F.

THINK OF IT A triad is plain toast; the 7th is the topping: same bread, new taste.

TRAP

- × Writing C7 when you mean Cmaj7: "7" alone means the bluesy one, with B♭.
- × Stacking the 7th from the wrong note: count a 7th up from the root, not from the 5th.

SEE ALSO
Four kinds of triad p. 70 · Chords in a key p. 72

PRACTISE IT
Workbook Harmony 5 · Online: Library › Chords

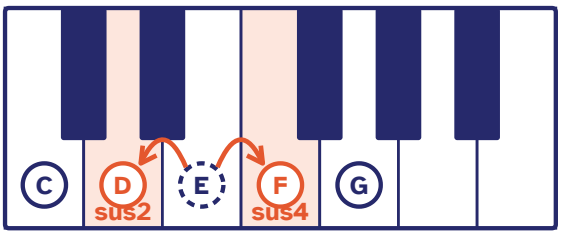
PART IV · HARMONY · 6

Colour chords

The 3rd is the chord's mood: major or minor, happy or sad. Slide it aside, take it away, or add one note, and a plain chord takes on a new colour. Five colours, all from C E G.

THE SECRET

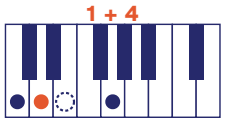
Move the 3rd, take it away, or add one note: five new colours.



1. Start from **C E G**. The **E** (the 3rd) makes it major.
2. Slide the E up to the next letter, **F**: **Csus4**.
3. Or down to the letter below, **D**: **Csus2**. With no 3rd, the chord is neither major nor minor: it **hangs in the air**.

Five colours, all on C

Csus2

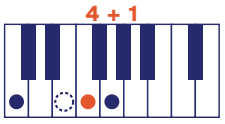


C D G

open

the 3rd steps down

Csus4

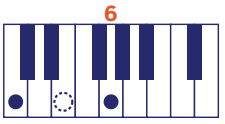


C F G

floating

the 3rd steps up

C5

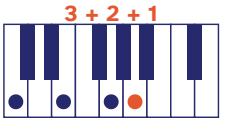


C G

strong

no 3rd at all

C6

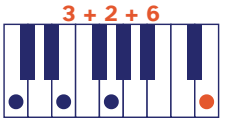


C E G A

warm

add the 6th

Cadd9

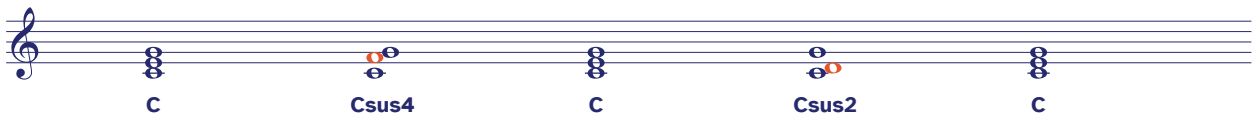


C E G D

sparkly

add the 9th on top

Sus wants to come home



The red note leans, then slides back to E: F steps down, D steps up. Play it slowly: C, Csus4, C, Csus2, C. Guitarists love this shimmer.

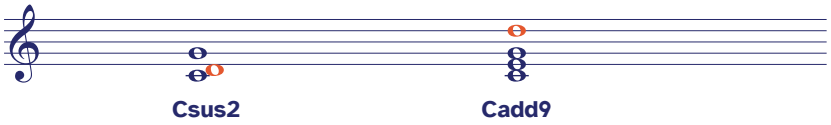
On any root, the counts stay the same

COLOUR	KEYS BETWEEN	ON C	ON G	ON F	ON D
sus2	1 + 4	C D G	G A D	F G C	D E A
sus4	4 + 1	C F G	G C D	F B♭ C	D G A
5	6	C G	G D	F C	D A
6	3 + 2 + 1	C E G **A**	G B D **E**	F A C **D**	D F♯ A **B**
add9	3 + 2 + 6	C E G **D**	G B D **A**	F A C **G**	D F♯ A **E**

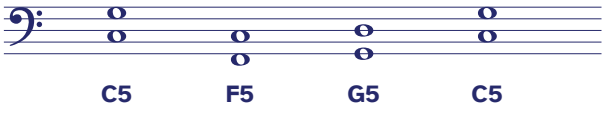
PART IV · HARMONY · 6 · CONTINUED

sus: the 3rd held up or down · **5**: no 3rd at all · **6** and **add9**: one note added on top. Csus means Csus4.

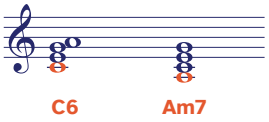
Spot the difference



The same red D, two jobs. Csus2 throws the E out; Cadd9 keeps the E and puts the D on top.



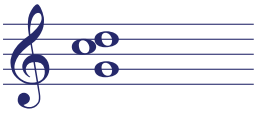
The power chord: just the root and the 5th, played low in the left hand. With no 3rd it is neither happy nor sad, only **strong**: the sound of rock guitar. Play C5, F5, G5, C5 and drum the rhythm.



C6 or Am7? The same four notes, C E G A. The red bottom note decides: C at the bottom is **C6**; A at the bottom is **Am7**. More on the bass note in Slash chords, p. 82.

EXAMPLE

Which colour is it? G C D



1. **Count the keys between:** G to C is 4, C to D is 1.
2. **4 + 1** is the sus4 card: no B (no 3rd), and C has taken its place.
3. **The root is G:** Gsus4.

G C D is **Gsus4**. Let the C fall to **B** and you have G major.

THINK OF IT Dressing the snowman: a scarf (the 6th) or a hat (the 9th) and it is still the same snowman. Sus moves its middle ball; a power chord takes the middle ball away.

TRAP

- × Writing C2: some books mean Csus2, others Cadd9. Check whether the E is there.
- × Calling Cadd9 a ninth chord: C9 also has the 7th (B♭). Add9 is just the triad plus the 9th (D on C), nothing else (The tower, p. 80).

SEE ALSO
Four kinds of triad p. 70 · Seventh chords p. 76

PRACTISE IT
Workbook Harmony 6 · Online: Library › Chords

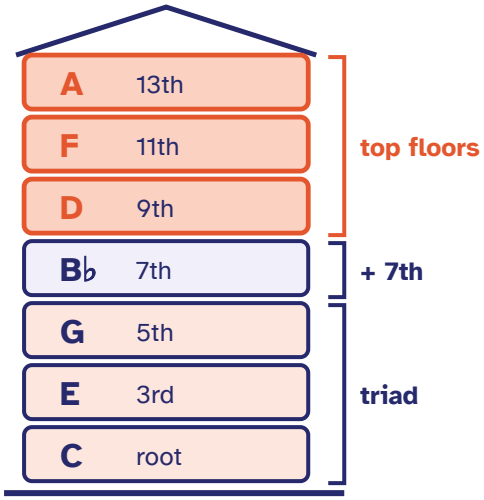
PART IV · HARMONY · 7

The tower: 9, 11, 13

Why stop at the 7th? Keep stacking 3rds and the building grows three more floors: the 9th, the 11th and the 13th. At the top you have used **all seven letters, A to G**.

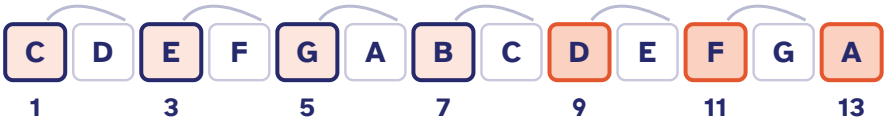
THE SECRET

Keep skipping letters above the 7th: 9, 11, 13 are 2, 4, 6 an octave up.



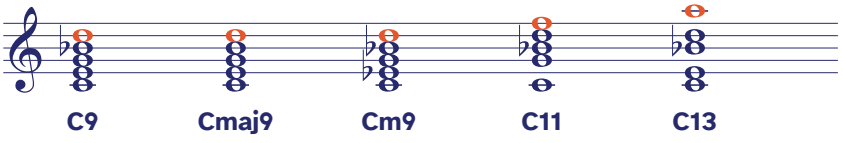
1. Build the seventh chord: **C E G B♭** (C7).
2. Keep skipping one letter: **D**, **F**, **A**. Those floors are the **9th, 11th and 13th**.
3. Take 7 away to find the letter: 9 → 2nd (D), 11 → 4th (F), 13 → 6th (A). Same letters, an octave up.

Skip a letter, all the way up



Start on C and take every other letter: C E G B D F A (letters only: in C7 the B is B♭). After the 13th comes C again: the tower is full.

Five tall chords on C

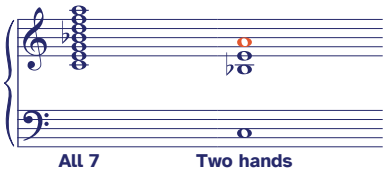


The red note is the top floor that gives each chord its name. C11 and C13 leave some notes out: see p. 81.

CHORD	BUILT AS	NOTES	SOUNDS	HEARD IN
C9	C7 + 9th	C E G B♭ **D**	soulful	funk, soul
Cmaj9	Cmaj7 + 9th	C E G B **D**	dreamy	film, ballads
Cm9	Cm7 + 9th	C E♭ G B♭ **D**	smooth	neo-soul, R&B
C11	C9 + 11th, no 3rd	C G B♭ D **F**	airy	gospel, pop
C13	C7 + 13th	C E B♭ (D) **A**	rich	jazz, blues endings

PART IV · HARMONY · 7 · CONTINUED

Too many notes? Leave some out



All seven notes sound muddy. Two hands play C13 with four: C in the left hand, B♭ E A in the right.

Ten fingers, seven floors: you choose. **Keep** the 3rd and the 7th (they say what kind of chord it is) and the red top note (its colour). **Leave out** the 5th first. The right hand can drop the root too, when the left hand or a bass player plays it.

With a major 3rd, the 11th (F) rubs against E. So C13 leaves out the F. C11 does the opposite: it keeps the F and leaves out the E.

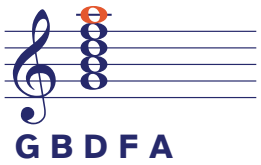
6 or 13? add9 or 9?

SAME TOP NOTE	WITHOUT THE 7TH	WITH THE 7TH (B♭)
A on C	**C6**: C E G A	**C13**: C E B♭ A
D on C	**Cadd9**: C E G D	**C9**: C E G B♭ D

Look for the 7th. No 7th: a colour chord (p. {H6}). A 7th: a tower chord.

EXAMPLE

Name this tall chord: G B D F A



1. **Find the root:** each note skips one letter, so the bottom note, G, is the root.
2. **The first four floors**, G B D F, count 3 + 2 + 2: G7.
3. **One more floor:** A is the 2nd above G, an octave up: the 9th.

G B D F A is **G9**: G7 with the 9th on top.

THINK OF IT A tower block has a lift: you don't stop at every floor. Get out at the 3rd, the 7th and the top floor with the view.

TRAP

- × Hunting far up for the 9th: it is just the 2nd, an octave higher (D on C).
- × Calling C13 a C6: the 13th chord has the 7th (B♭) as well.
- × Playing F and E together in C13 or C11: they clash. C13 drops the F; C11 drops the E.

SEE ALSO
Seventh chords p. 76 · Colour chords p. 78

PRACTISE IT
Workbook Harmony 7 · Online: Library › Chords

PART IV · HARMONY · 8

Slash chords

C/E means one chord with a chosen bass note: play a C chord, with E as the lowest note. It is not a fraction, and not two chords.

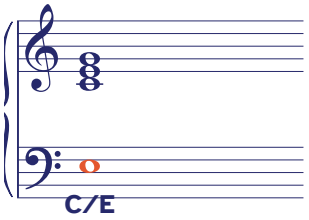
THE SECRET
Chord before the slash, bass note after it.

C/E

the chord
right hand

the bass
left hand, lowest note

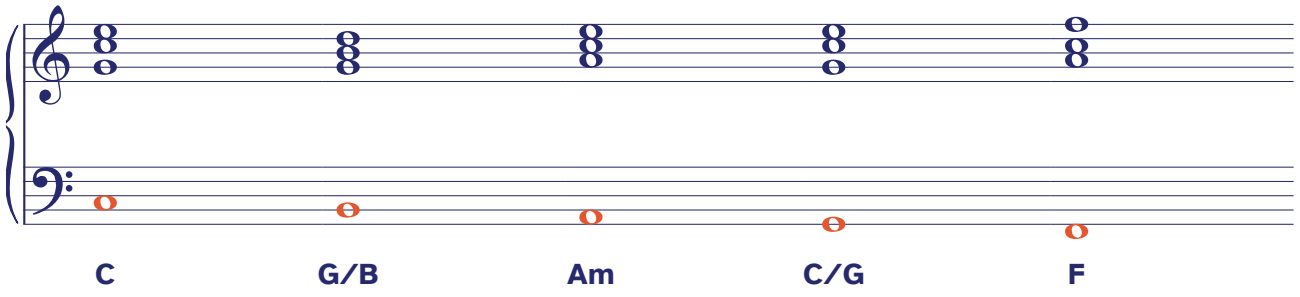
1. Before the slash: the **chord**. C means C E G, in the right hand.
2. After the slash: the **bass**. E goes in the left hand, below everything.
3. Play them together: **C/E**, “C over E”.



Four jobs for a slash chord

JOB	EXAMPLE	THE BASS NOTE IS	WHAT YOU HEAR
Inversion	C/E · C/G	a note of the chord	smoother changes (p. 74)
Walk-down	C · G/B · Am	one step down each time	a bass line that walks down
Pedal	F/C · G/C	the same note, held	chords move over a floor
Colour	F/G · D/C	not in the chord	open, modern: pop, gospel

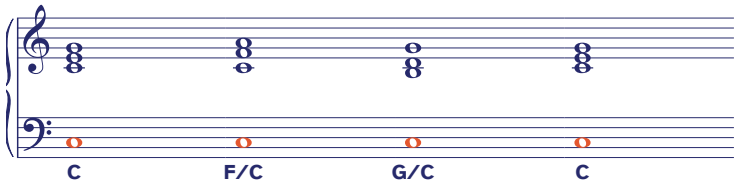
Job 2: a bass that walks down



The red bass walks straight down, C B A G F, while the right hand moves a step at most. G/B and C/G make the steps possible.

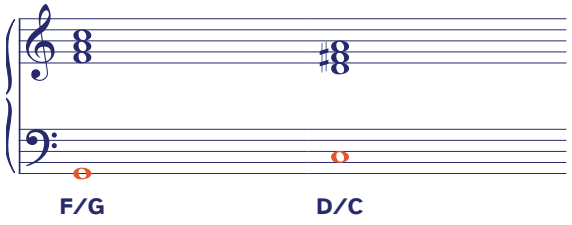
PART IV · HARMONY · 8 · CONTINUED

Job 3: a pedal, one bass under everything



The bass holds C while the chords change above it: C, F/C, G/C, C. The chords pull away from the C, then come home. (It is called a pedal after the organ's foot pedals, which hold long low notes.)

Job 4: a bass from outside the chord

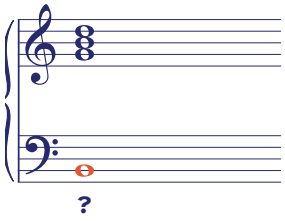


F/G: F A C over a G. G is not in the F chord, so it floats, open and unfinished, ready to land on C. Pop and gospel players love it.

D/C: D F♯ A over a C. The C is not in the D chord, so it rubs, then usually steps down to B (D/C to G/B).

EXAMPLE

Name it: G B D over a low B



1. **The right hand:** G B D, the G chord.
2. **The lowest note:** B, in the left hand.
3. **Chord, slash, bass:** G/B. B is in the G chord: an inversion.

It is **G/B**, "G over B": G major, 1st inversion.

THINK OF IT A house and the ground it stands on: the slash says what to build (the chord) and where to build it (the bass). Same house, new ground, new view.

TRAP

- × Writing it back to front: the chord comes first, the bass second. E/C would be an E chord over C.
- × Letting the right hand dip below the bass: the letter after the slash must be the lowest note you play.
- × Forgetting the chord itself: in C/E the right hand still plays the whole C chord, C E G, not just C.

SEE ALSO
Inversions p. 74 · Colour chords p. 78

PRACTISE IT
Workbook Harmony 8 · Online: Library › Chords

PART IV · HARMONY · 9

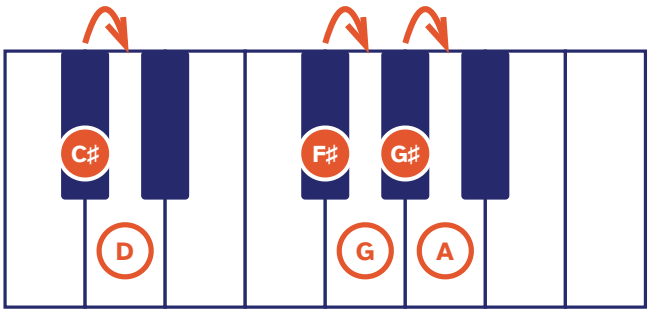
Chords from outside the key

C major lives on the white keys. A black key in a chord is a **guest**. Guests never stay: each one moves **one key**, and shows you where the music goes next.

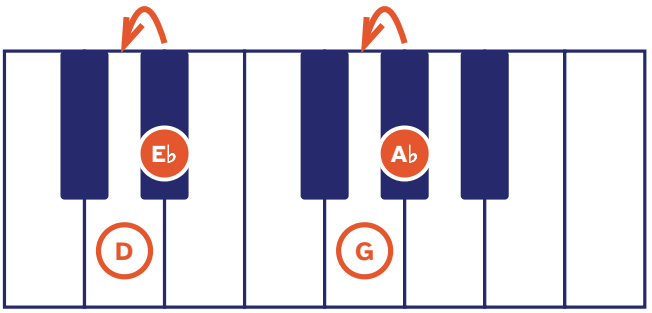
THE SECRET

Sharp guests push up, flat guests sigh down: follow them one key and you find the next chord.

Sharp guests push up

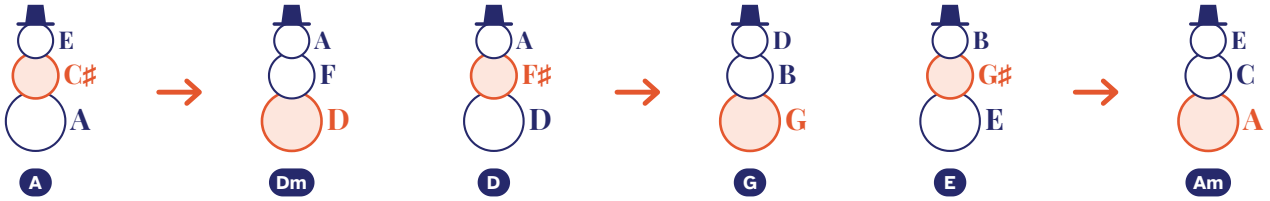


Flat guests sigh down

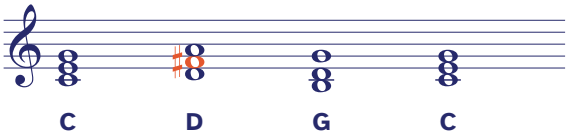


1. Find the **black key** in the chord: that is the guest.
2. A **sharp**? Go **up** one key. A **flat**? Go **down** one key.
3. You have found a note of the **next chord**. Play it and hear it land.

Sharp guests: up one key names the next chord

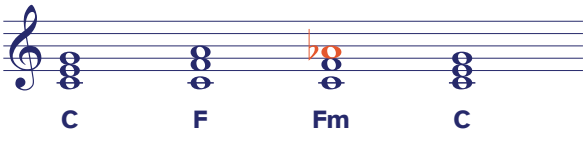


C♯ goes up to **D**, so the next chord is D minor. F♯ goes up to **G**: G. G♯ goes up to **A**: A minor.



Try it in a song: C, D, G, C. The **F♯** pushes up into G, then G leads home to C.

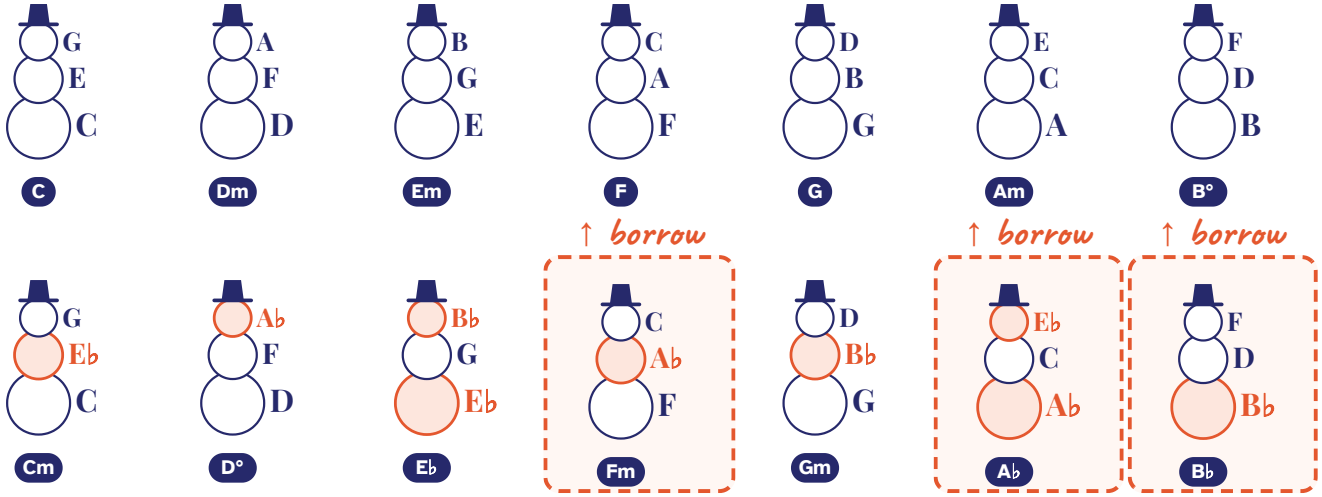
Flat guests: borrowed from the twin, they sigh down



C major and C minor are **twins**: the same home note, C, in different clothes. A flat guest is one piece of the twin's outfit. In C, F, Fm, C the **A♭** sighs down to G.

PART IV · HARMONY · 9 · CONTINUED

C major *at home*

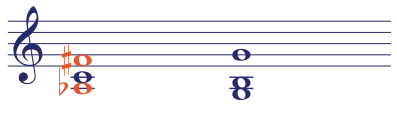


C minor *the twin: same home note, darker clothes*

Their names, for later

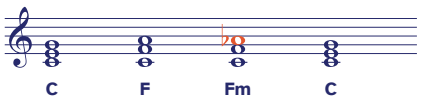
GUEST	NOTES	MOVES	NEXT CHORD	ITS NAME
sharp guest	D F♯ A	F♯ up to G	G	V of V
sharp guest	A C♯ E	C♯ up to D	Dm	V of ii
sharp guest	E G♯ B	G♯ up to A	Am	V of vi
flat guest	F A♭ C	A♭ down to G	C	borrowed iv
flat guest	A♭ C E♭	A♭ down to G	G or B♭	borrowed ♭VI
flat guest	B♭ D F	B♭ on to C	C	borrowed ♭VII

B♭ is the one flat guest that walks **up** instead, on to C: the rock and film ending B♭ → C. The bigger guests (the Neapolitan, Level 7) follow the same idea.



An advanced guest (Extra): the augmented 6th has two guests at once. The flat **A♭** falls and the sharp **F♯** rises, and both land on G. The rule still works.

TRY IT · 10 SECONDS



Play C, F, Fm, C slowly.

Hear the A♭ sigh down to G? Now play C, D, G, C: the F♯ pushes up instead. One sighs, one pushes.

PART IV · HARMONY · 9 · CONTINUED

EXAMPLE

Find the guest: A C♯ E in C major



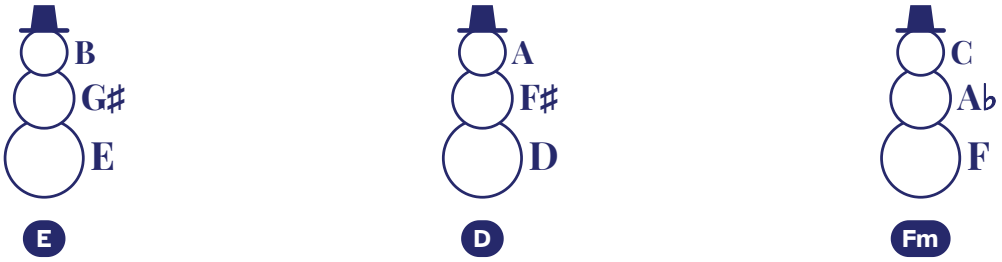
A C♯ E

1. **The black key:** C♯. That is the guest.
2. **It is a sharp,** so it goes up one key: to D.
3. **D names the next chord:** D minor (D F A).

A C♯ E leads to **D minor**. Play them one after the other and hear C♯ slide into D.

GAME

Where does the guest go?



In C major: find each guest, follow it one key, and write the chord that comes next.

Answer: 1 G♯ up to A: A minor · 2 F♯ up to G: G · 3 A♭ down to G: C.

THINK OF IT A surprise guest at dinner: one new face changes the whole conversation, and then everyone goes home.

TRAP

- × Following a sharp downwards: sharps push **up**, flats sigh **down**.
- × Learning the names first: play it, hear where the guest goes, then name it.
- × Mixing up the relatives: C minor is C major's twin (same home note). A minor is its neighbour (same key signature). We borrow from the twin.

SEE ALSO
Chords in a key p. 72 · Slash chords p. 82 · Cadences p. 87

PRACTISE IT
Workbook Harmony 9 · Online: Library › Chromatic chords

PART IV · HARMONY · 10

Cadences

A phrase is a musical sentence, and it needs punctuation. The last two chords of a phrase are its **cadence**. They tell you: finished, taking a breath, or surprise!

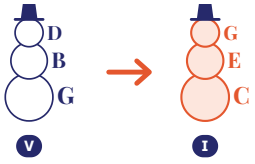
THE SECRET

Music has full stops and commas too: listen to the last two chords of a phrase.

.

Perfect

V → I

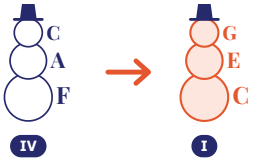


Home. The end.

Amen

Plagal

IV → I

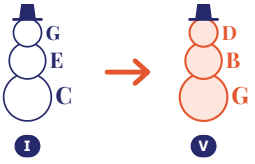


A soft, gentle end.

,

Imperfect

… → V

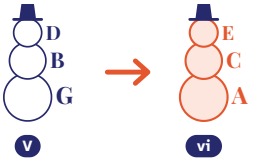


To be continued…

?!

Interrupted

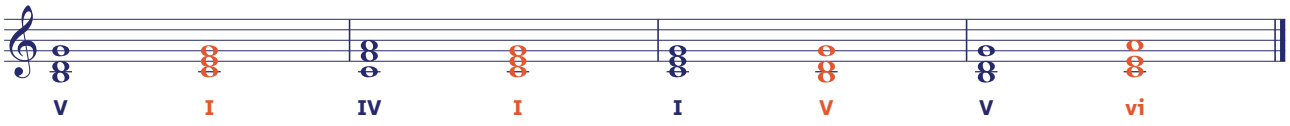
V → vi



Surprise! Not home.

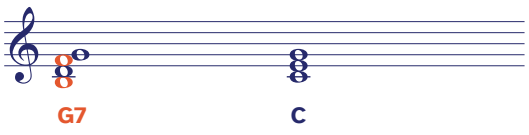
1. Find the end of a phrase: often a long note, or a breath.
2. Name its **last two chords** with numerals: V I, IV I, anything then V, or V vi.
3. Match the punctuation: full stop (.), Amen, comma (,) or ?!

The four cadences in C



Perfect, plagal, imperfect, interrupted. The red chord is where each one lands.

CADENCE	CHORDS	FEELS LIKE	ALSO CALLED
Perfect	V → I	a full stop: finished	authentic, full close
Plagal	IV → I	the “A-men” sung at the end of a song in church	Amen cadence
Imperfect	any → V	a comma: waiting	half close, half cadence
Interrupted	V → vi	a surprise: not home yet	deceptive

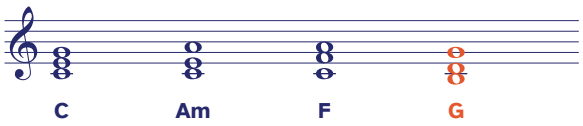


Make the full stop stronger: add the 7th to V. In G7 the **F** falls to E and the **B** rises to C: two small steps home (Seventh chords, p. 76).

PART IV · HARMONY · 10 · CONTINUED

EXAMPLE

How does this phrase end? C Am F G



1. **The last two chords:** F, then G.
2. **As numerals:** IV, then V.
3. **It lands on V:** the comma.

An **imperfect cadence**: the music takes a breath, and the next phrase answers it.

GAME

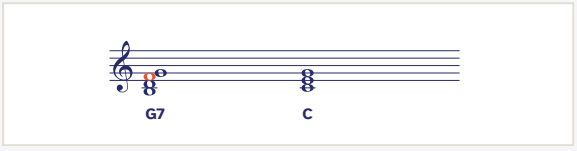
Punctuate the phrases



Each pair ends a phrase in C major. Turn the letters into numerals first (C is I, F is IV, G is V, Am is vi), then write full stop, comma, Amen or ?! beside each.

Answer: 1 Amen (plagal), 2 ?! (interrupted), 3 comma (imperfect), 4 full stop (perfect).

TRY IT · 10 SECONDS



Play G7, then C. The end!

Now play G7, then A minor: “huh?!” That is the interrupted cadence. Then play F, then C, and sing “A-men”.

THINK OF IT Punctuation in a sentence: a full stop ends it, a comma makes you wait, and a surprise makes you gasp. Music ends its phrases the same way.

TRAP

- × Calling every ending on I (C) perfect: a perfect cadence comes **from V**. F to C is plagal.
- × Mixing up imperfect and perfect: imperfect **ends on** V; perfect **starts on** V and lands on I.

SEE ALSO

Chords in a key p. 72 · Seventh chords p. 76 · Chords under a melody p. 89

PRACTISE IT

Workbook Harmony 10 · Online: Library › Cadences

PART IV · HARMONY · 11

Chords under a melody

The melody is the singer; the chords are the band. To choose a chord for each bar, look at the melody's main notes and pick the snowman holding them.

THE SECRET

Look at the melody's main notes, then pick the snowman that holds them.



I



IV



V
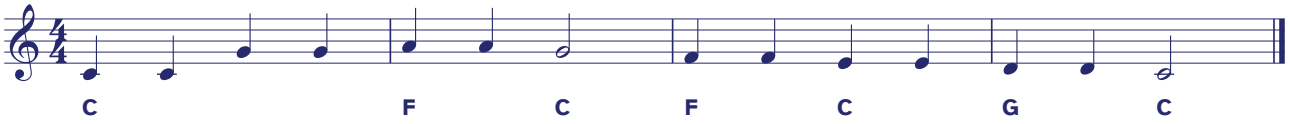
C	D	E	F	G	A	B
I, IV	V	I	IV	I, V	IV	V

Which snowman holds the tune's main notes?

Every note of C major lives in I, IV or V. C and G live in two: let your ear choose.
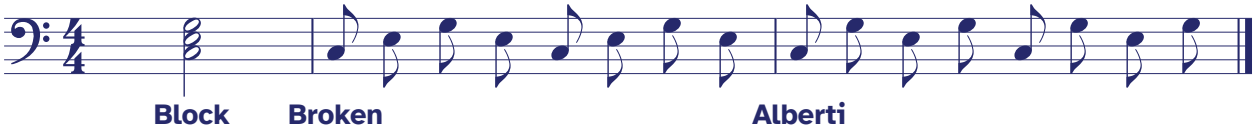
1. Look at the **main notes** of the bar: the long ones, and the ones on beats 1 and 3.
2. Find the snowman that holds them: **I, IV or V**.
3. Play that chord in the left hand while the right hand plays the tune.

Twinkle, Twinkle, with its chords



C C G G sit in I (C); A A in IV (F), then G in I; F F in IV, E E in I; D D in V (G), and the last C in I. The chord changes halfway through bars 2, 3 and 4.

Three ways to play the chord in the left hand



Block: all three at once. Broken: one at a time, up and down. Alberti: low, high, middle, high, named after Domenico Alberti, who loved it.

PART IV · HARMONY · 11 · CONTINUED

EXAMPLE

Choose the chord: E D C D | E E E

E D C D | E E E

red = beats 1 and 3

1. **Bar 1:** the main notes are E and C (beats 1 and 3).
2. **E and C** both live in I.
3. **Bar 2:** E E E lives in I too.

Both bars take **I** (C). That is the start of Mary Had a Little Lamb.

GAME

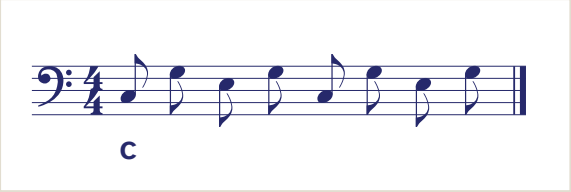
Pick the snowman

F A F A	D B G B	G E C E
1	2	3

Each bar is in C major. Which snowman holds its notes: I, IV or V?

Answer: 1 IV (F A C) · 2 V (G B D) · 3 I (C E G).

TRY IT · 10 SECONDS



Left hand: C G E G, over and over.

That is the Alberti bass on the C chord. Keep it going while you hum Twinkle, then try it on F (F C A C) and G (G D B D).

YOU KNOW THIS

Mozart, Sonata in C, K. 545: the left hand



The opening left hand is an **Alberti bass**: C G E G is just the C chord, broken into low, high, middle, high; D G F G is G7 (G B D F, without the B). Mozart wrote it an octave higher, starting on middle C.

Try it. Play this left hand alone, then add the famous tune: C, E G, B, C D C.

THINK OF IT Like picking a coat for the weather: when the tune changes, change the chord.

TRAP

- × Choosing a chord for a passing note: use the main notes (strong beats, long notes).
- × Playing the left hand louder than the tune: the melody is the star, the chords are the stage.

SEE ALSO

Chords in a key p. 72 · Cadences p. 87

PRACTISE IT

Workbook Harmony 11 · Online: Library › Chords

PART IV · HARMONY · 12

Progressions everywhere

A **progression** is a trip from chord to chord. A handful of trips turn up in song after song. Learn them by their numbers and you can play them in any key.

THE SECRET
Most songs take the same few trips: home, away, a pull, and home again.

I home · vi iii home's cousins · IV ii away · V the pull back home

The pop four
countless pop songs and ballads

I C → V G → vi Am → IV F

The fifties
old rock 'n' roll and doo-wop

I C → vi Am → IV F → V G

The jazz three-step
jazz standards (as ii7, V7, Imaj7)

ii Dm → V G → I C

The canon
Pachelbel's Canon and its many children

I C → V G → vi Am → iii Em → IV F → I C → IV F → V G

1. Learn the trip by **numbers**: I V vi IV.
2. Pick a key and turn the numbers into chords: in C, **C G Am F**.
3. Play it round and round. Change key: the numbers stay the same.

Same trip, any key

The pop four	I	V	vi	IV
in C	C	G	Am	F
in G	G	D	Em	C
in F	F	C	Dm	B♭
in D	D	A	Bm	G

EXAMPLE

Play the pop four in G major

I G → V D → vi Em → IV C

1. **I** is the key note: G.
2. **V** is the 5th note of the G scale: D. **vi** is the 6th: E, minor, so Em.
3. **IV** is the 4th note: C.

G, D, Em, C: **the pop four in G**. Same trip, new key.

PART IV · HARMONY · 12 · CONTINUED

GAME

Name the trip

1 C · Am · F · G

2 Dm · G · C

3 C · G · Am · F

All in C major. Turn each into numbers, then name the trip: the pop four, the fifties or the jazz three-step.

Answer: 1 the fifties (I vi IV V) · 2 the jazz three-step (ii V I) · 3 the pop four (I V vi IV).

TRY IT · 10 SECONDS
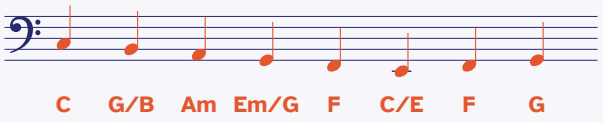


Play C, G, Am, F, four beats each.

Go round four times without stopping, left hand plays the chord's name note (C for C, A for Am), right hand the whole snowman. Hum any tune you like over it.

YOU KNOW THIS

Pachelbel's Canon: the trip under the tune



The famous bass walks down, and above it the chords take the **canon trip**: C G Am Em F C F G. Countless songs borrow it.

Try it. Play the bass slowly with your left hand and the chords in your right.

THINK OF IT A walk round the block: leave home, wander, feel the pull, and turn the corner back home.

TRAP

- × Learning songs one by one: learn the trip by numbers, then change key.
- × Mixing up the pop four and the fifties: same four chords, different order. Listen to which chord comes second.

SEE ALSO

Chords in a key p. 72 · Cadences p. 87 · Chords under a melody p. 89

PRACTISE IT

Workbook Harmony 12 · Online: Library › Chords

PART IV · HARMONY · TIPS

When a chord sounds wrong

A chord that sounds wrong usually has one small slip. Name the slip, fix that one note, and the chord comes right.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A chord sounds sour	**the kind**: keys between, 3 + 2 or 2 + 3	p. 70
Your hand jumps all over	**an inversion**: keep the shared note still	p. 74
A big chord sounds muddy	**leave notes out**: drop the 5th first	p. 80
A sharp or flat surprises you	**a guest**: sharps go up, flats go down	p. 84
The end does not feel finished	**the cadence**: end with V → I	p. 87
The left hand fights the tune	**the main notes**: pick the snowman holding them	p. 89

Four ways to learn chords fast

1 Build, then play
Say the snowman (C E G) before your fingers go down.

2 Hands alone first
Left hand chords alone, then the tune alone, then together.

3 One trip a week
Learn one progression, then play it in three keys.

4 Listen for the kind
Play major and minor side by side: bright, then dark.

EXAMPLE

Fix a sour chord



1 **It sounds dark**, but the music wants a bright C chord.

2 **Count the keys between:** 2, then 3. That is minor.

3 **Move one note:** E♭ up to E makes 3 + 2. Bright again.

One note was off: **fix the middle of the snowman**.

THINK OF IT Like a mechanic: one odd sound points to one part. Fix that part and the engine runs smoothly.